Practical Curriculum Design: Security, Automation, & IoT Systems

Total Course Duration: 16 Weeks (80 Days)

Time Commitment: 5 Days/Week, 2 Hours/Day (Total: 160 Hours)

Delivery Model: Practical-Oriented (30% Concepts/Theory, 70% Hands-On Labs)

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1. Overall Module Weightage & Time Allocation

To ensure students are industry-ready, the curriculum is divided into 4 Core Modules plus dedicated capstone Practical Labs. The time allocation is weighted based on the complexity of installation, programming requirements, and market demand.

Module	Duration	Total Hours	Weightage (%)	Primary Focus
Module 1: Electrical & Networking	3 Weeks	30 Hours	18.75%	Safe power wiring, network configuration, cabling
Module 2: Professional CCTV Systems	3.5 Weeks	35 Hours	21.875%	Camera selection, storage design, analytics, IP configuration
Module 3: Security & Building Automation	3.5 Weeks	35 Hours	21.875%	Commercial security systems, fire alarms, access control
Module 4: Smart Home, IoT & Business	4 Weeks	40 Hours	25.00%	Home protocols, system integration, entrepreneurship, sales
Professional Practical Labs (Capstone)	2 Weeks	20 Hours	12.50%	Integrated system deployment, splicing, troubleshooting
Total	**16 Weeks**	**160 Hours**	**100%**	**Comprehensive practical competency**

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2. Detailed Topic Weightage & Hours Breakdown

Below is the recommended weightage for each topic within the modules, structured as compact tables indicating weeks (Wk), hours (Hrs), sessions (Sess), practical focus, and detailed theory/PPT slide bullet points.

Module 1: Electrical & Networking (30 Hours)

This module establishes the foundational hardware and network infrastructure.

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W1	**Electrical Fundamentals (AC/DC)**	2 Hrs	1.0	Voltage, current, resistance, series/parallel circuits.	- Ohm's Law (V=IR), Power Formula (P=VI) - AC vs DC: sine wave, frequency, polarization differences - Single-phase (230V) vs 3-phase (415V) systems

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W1	**Electrical Safety & PPE**	1 Hr	0.5	Lockout/Tagout (LOTO), working with live currents safely.	- Hazards of electric shock, burns, and arc flashes - PPE classes: insulated gloves, safety shoes, insulated tools - LOTO procedure steps & emergency shock response
W1	**Earthing & Lightning Protection**	2 Hrs	1.0	Grounding concepts, surge protection devices (SPDs).	- Purpose of earthing: low resistance path for fault currents - Types: Plate, Pipe, and Chemical Earth pits - SPDs: Clamping surge voltages, Lightning rod zones
W1	**Reading Electrical Drawings**	2 Hrs	1.0	Schematic symbols, single-line diagrams (SLDs).	- Standard CAD symbols for relays, switches, fuses, and load - Interpreting wiring schematics vs single-line diagrams - Tracking wire markers & color-coding standards
W1	**Cable Selection & Management**	2 Hrs	1.0	Sizing wires, conduit bending, raceways.	- Wire gauge metrics (AWG vs Sq mm) & current ratings - Voltage drop calculations over long distances - Conduits: PVC vs GI, fill capacity rules, bending standards
W1/2	**Power Supply & SMPS**	2 Hrs	1.0	Checking output voltage, current ratings, backup batteries.	- Switched-Mode Power Supply (SMPS) working principle - Selecting ratings: voltage regulation, peak current capacity - Battery sizing: Ah calculations, load vs backup time
W2	**Multimeter Practical**	3 Hrs	1.5	Testing voltage, current, resistance, and continuity of components.	- Multimeter dials: AC/DC voltage, Amp, Ohm, Diode settings - Measuring parameters in series vs parallel circuits - Testing switches, checking fuse continuity, detecting shorts
W2	**IP Addressing & Subnetting**	4 Hrs	2.0	IPv4, subnet masks, gateway configurations, DHCP vs Static.	- IPv4 Structure: Network vs Host ID, Class A, B, C divisions - Subnetting & CIDR notation (e.g., /24, /28, /30 masks) - Gateway, Subnet, and Public vs Private IP ranges (RFC 1918)
W2/3	**Router & Switch Configuration**	4 Hrs	2.0	WAN/LAN setup, port forwarding, basic Wi-Fi configs.	- Managed vs Unmanaged switches, routing tables overview - NAT (Network Address Translation) & Port Forwarding - Basic Wi-Fi security standards (WPA2, WPA3, channel conflict)
W3	**VLAN Basics**	2 Hrs	1.0	Segmenting network traffic (IP Cameras vs Data network).	- Purpose of VLANs: segregate broadcast domains, security - Tagging standard: IEEE 802.1Q - Access ports (single VLAN) vs Trunk ports (multi-VLAN)

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W3	**PoE & PoE Switches**	2 Hrs	1.0	PoE standards (802.3af/at/bt), power budget calculations.	- PoE standards: 802.3af (15.4W), 802.3at (PoE+ 30W), 802.3bt (90W) - Active vs Passive PoE; calculating PoE switch power budget - Max transmission distance limitations (100m rule)
W3	**Fiber Optic Basics**	2 Hrs	1.0	Multi-mode vs Single-mode, media converters, SFP modules.	- Optical propagation: Total Internal Reflection - Single-mode (long distance, laser) vs Multi-mode (LED) - Connector types (LC, SC, ST) & SFP transceiver modules
W3	**Network Troubleshooting**	2 Hrs	1.0	IP conflicts, ping test, trace route, cable testers.	- Command-line tools: ping, traceroute/tracert, ipconfig, nslookup - Resolving duplicate IP addresses, packet loss causes - Diagnostic tools: Wiremap testers, tone generator/probe

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Module 2: Professional CCTV Systems (35 Hours)

CCTV is one of the highest-demand skills. This module balances video analytics design with hardware setup.

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W4	**Analog & IP CCTV**	2 Hrs	1.0	Architecture comparison (Coaxial vs Cat6 topologies).	- Coaxial-based Analog (HD-TVI, CVI, AHD) vs IP topologies - Latency, resolution limits, power requirements comparison - Hybrid migration paths (Analog to IP)
W4	**Camera Selection & Design**	3 Hrs	1.5	Form factors, lux levels, WDR, placement planning.	- Form factors: Bullet, Dome, Turret, Eyeball, PTZ - Sensor metrics: Lux levels (low light sensitivity), WDR (dB) - Planning criteria: DORI standard (Detect, Observe, Recognize, Identify)
W4	**Lens & Field of View**	2 Hrs	1.0	Focal length calculations, varifocal vs fixed, calculator tools.	- Focal length (mm) vs Angle/Field of View (FoV) - Fixed lens vs Varifocal/Motorized zoom lens differences - Using lens calculator apps for precise coverage planning
W4	**DVR, NVR & Hybrid Systems**	3 Hrs	1.5	Hardware channels, PoE NVR configurations, HDD installation.	- NVR (Network Video Recorder) vs DVR (Digital Video Recorder) - Incoming/Outgoing bandwidth limits (Mbps capacity) - Internal PoE ports vs Uplink network ports
W5	**RAID & Storage Design**	2 Hrs	1.0	RAID configurations (0/1/5/10), storage capacity math.	- RAID levels: RAID 0, 1, 5, 10 (speed, redundancy, cost) - Calculating HDD capacity: Resolution, Frame rate, Bitrate, Days - Selection of Surveillance-grade HDDs vs standard Desktop HDDs

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W5	**ONVIF Standards**	1 Hr	0.5	Profile S/G/T compliance and cross-brand compatibility.	- What is ONVIF: open industry forum standard - Profile S (basic video), G (edge storage), T (advanced streaming/metadata) - Connecting third-party cameras to NVRs using ONVIF
W5	**Video Compression (H.264/H.265/H.265+)**	2 Hrs	1.0	Bandwidth impact, frame rates, bitrate control (CBR vs VBR).	- Codecs: H.264 vs H.265 vs H.265+ (Smart codecs) - Frame rates (FPS) vs Resolution vs Bitrate - Constant Bitrate (CBR) vs Variable Bitrate (VBR) applications
W5	**AI Analytics**	2 Hrs	1.0	Line crossing, intrusion detection, people counting configuration.	- Rules: Line crossing, intrusion zones, object left/removed - Smart searches: querying metadata (human/vehicle classifications) - People counting setup, calibration, and report generation
W5	**ANPR (Automatic Number Plate Recognition)**	2 Hrs	1.0	Camera angle, vehicle speed settings, database OCR.	- ANPR camera positioning: horizontal angle <30°, vertical angle <30° - Shutter speed settings for high speed vehicle capture (>1/1000s) - Whitelist/blacklist action triggers (barrier open, alert)
W5/6	**Facial Recognition**	2 Hrs	1.0	Database creation, whitelist/blacklist rules, alert parameters.	- Enrollment requirements: Face size (pixels), illumination, angles - Matching thresholds: False Acceptance vs False Rejection rates - Integrating facial recognition triggers with relay systems
W6	**ColorVu, AcuSense & Thermal Cameras**	3 Hrs	1.5	Dual-light setups, false alarm reduction, thermal imaging.	- ColorVu (F1.0 super aperture, warm supplementary light) - AcuSense (deep learning classification to filter false alarms) - Thermal: long-range perimeter protection, fire point detection
W6	**PTZ Cameras**	2 Hrs	1.0	Preset coordinates, patrol tours, auto-tracking setup.	- PTZ components: Pan, Tilt, Zoom mechanisms, optical magnification - Creating Presets, Patrols, and Patterns - Auto-tracking logic (using deep learning to follow targets)
W6	**Remote Viewing & Cloud**	3 Hrs	1.5	P2P connections, DDNS, mobile apps, VMS software.	- Cloud P2P connection working model (no port forwarding required) - DDNS (Dynamic DNS) setups for dynamic IP networks - VMS (Video Management Software) vs Web client vs Mobile apps
W6/7	**Cybersecurity**	2 Hrs	1.0	Changing default ports, complex password policy, disabling RTSP.	- Disabling default credentials; enforcing complex passwords - Disabling UPnP, change standard ports (HTTP, RTSP, SDK) - Restricting access via IP filters, HTTPS certificates

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W7	**Troubleshooting & Maintenance**	4 Hrs	2.0	No-video diagnostics, IP clashes, low night-vision, lens cleaning.	- Debugging "No Video" / "Network Offline" issues - Resolving IR reflection (halo effect) due to dirty dome glass - Focus calibration, focus drift fixing, firmware upgrades

Module 3: Security & Building Automation (35 Hours)

This module covers commercial low-voltage (ELV) systems.

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W7	**Video Door Phone**	2 Hrs	1.0	Multi-apartment vs villa systems, wiring indoor/outdoor monitors.	- 2-wire vs IP-based Video Door Phone (VDP) topologies - Door lock integration (NO/NC relay options, lock power) - Multi-apartment SIP servers & outdoor call station routing
W7	**Access Control**	3 Hrs	1.5	Wiegand controllers, card readers, exit switch wiring.	- Controller protocols: Wiegand vs OSDP (secure, bi-directional) - Reader connections, door contact sensors, exit button wiring - Controller configuration (access groups, time zones, anti-passback)
W8	**Biometric Systems**	2 Hrs	1.0	Fingerprint, facial, and RFID enrollment; standalone vs network models.	- Biometric technologies: Optical vs capacitive fingerprint, 3D face - Verification modes: Card ONLY, Card+PIN, Finger/Face ONLY - Network architecture: central server push vs local device database
W8	**Smart Locks**	2 Hrs	1.0	Electromagnetic (EM) locks, bolt locks, strike locks, cabinet locks.	- EM Lock (fail-safe: releases on power loss) vs Strike (fail-secure) - Holding force ratings (600 lbs, 1200 lbs) & bracket selection (L/Z/U) - Flyback diode application to prevent relay burn-out
W8	**Boom Barrier**	2 Hrs	1.0	Induction loop detectors, photo-beams, controller logic boards.	- Boom barrier components: Motor, springs, balance mechanism, logic board - Loop detectors (vehicle presence detection) & loop coil math - Safety sensors: Photo-electric beams, pressure strips
W8	**Gate Automation**	2 Hrs	1.0	Sliding/swing gate motors, actuator arms, remote control pairing.	- Sliding gate: pinion/rack drive, magnetic/mechanical limit switches - Swing gate: articulated arm vs linear actuator motors - RF remote control pairing: rolling codes vs fixed codes
W8/9	**Intruder Alarm Systems**	3 Hrs	1.5	Hardwired/wireless zones, PIR motion, magnetic door contacts, glass break.	- Panel architecture: zones (EOL resistors, tamper loops) - Sensor types: Dual-tech PIR (microwave + IR), glass break, contacts - Arming profiles: Arm Away, Arm Stay/Home, Panic/24-hour zones

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W9	**Fire Alarm Systems**	4 Hrs	2.0	Conventional vs Addressable systems, smoke/heat detectors, sounders.	- Conventional (zone voltage drop) vs Addressable (SLC loop protocols) - Detectors: Ionization/Photoelectric smoke, Thermal (ROR/Fixed) - Safety integrations: Lift homing, AHU trip, access doors release
W9	**PA/BGM Systems**	2 Hrs	1.0	100V amplifiers, ceiling/wall speakers, zone selection, LMT calculations.	- 100V line system working principle: step-up/down transformers - Impedance matching: calculating total speaker wattage vs amplifier load - Zone selectors, attenuators, paging microphone priority override
W9	**Intercom Systems**	2 Hrs	1.0	Analog PABX, IP-PBX, SIP endpoints, extension routing.	- Analog PABX trunk lines/extensions vs IP-PBX SIP trunks - IP intercom protocols (SIP), codec requirements (G.711/722) - Programming call groups, call forwarding, intercom rings
W9/10	**Elevator Access Control**	2 Hrs	1.0	Controller relay boards, restricted floor access configuration.	- Elevator integration: Dry contact relay output card setup - Restricting floors based on card credentials, scheduling controls - Emergency override integration (fire alarm integration)
W10	**Visitor Management**	2 Hrs	1.0	Software badges, kiosk sign-in, cloud database logging.	- Software database structure: Pre-registration, host approvals - Hardware peripherals: Badge printers, QR scanners, camera integrations - Privacy and data regulations (GDPR/local data storage rules)
W10	**Hotel Lock Systems**	2 Hrs	1.0	Keycard encoders, energy saver switches, PMS software integration.	- Card technologies: Mifare, RFID tags, offline card encryption - Energy saver switches (card presence detection for room power) - Property Management System (PMS) integration protocols
W10	**Parking Management**	2 Hrs	1.0	Ticket dispensers, card collectors, loop integration, barrier pairing.	- Ticket dispensers (barcode/thermal) & parking loop sequence logic - Auto card-collector systems for visitor exit validation - Cash register workstation integration, automated pay stations
W10	**Basic Building Management Systems (BMS)**	3 Hrs	1.5	DDC controllers, sensor calibration, BACnet/Modbus integration.	- DDC (Direct Digital Control) architecture & input/output types (AI/DI/AO/DO) - BMS protocols: BACnet IP/MSTP, Modbus RTU/TCP - Monitoring HVAC, lighting, and water meters

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Module 4: Smart Home, IoT & Business (40 Hours)

Smart Home integration requires high logical skill (programming systems), and establishing the business requires commercial execution.

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W11	**KNX Fundamentals**	4 Hrs	2.0	ETS software database config, bus line wiring, system devices.	- KNX standard: TP1 green bus line, decentralized topology - System devices: Power supply, choke, IP router, actuator, sensor - ETS (Engineering Tool Software): Group addresses, download logic
W11	**Wi-Fi Smart Home**	2 Hrs	1.0	Tuya, SmartLife ecosystem, addressing router client limit issues.	- Consumer Wi-Fi modules: Tuya, SmartLife, Sonoff ecosystems - Router limits (DHCP pool exhaustion, airtime fairness limits) - Smart switches: neutral wire vs no-neutral (capacitor) wiring
W11	**Zigbee & Z-Wave**	3 Hrs	1.5	Mesh network structures, coordinators, routers, low power metrics.	- Zigbee (2.4GHz) vs Z-Wave (sub-GHz) frequency comparison - Network roles: Coordinator (Hub), Routers (Mains powered), End Devices - Mesh healing, latency, sleep mode logic for battery life
W11/12	**Matter & Thread**	2 Hrs	1.0	Unified standard, thread border routers, ecosystem interop.	- What is Matter: Application layer standard running on IP - Thread: Low-power wireless mesh transport protocol (IPv6 based) - Multi-admin feature: controlling one device across Apple, Google, Alexa
W12	**Alexa & Google Home Integration**	2 Hrs	1.0	Cloud linking, custom voice routines, device pairing.	- Cloud-to-Cloud (OAuth linking) vs Local Home SDK controls - Creating Routines/Automations using variables (time, voice, sensor) - Voice profile matching, smart home skill development basics
W12	**Home Assistant**	4 Hrs	2.0	Local hub OS, YAML configuration, dashboards, advanced automations.	- OS installation types (HAOS on Pi/NUC, Docker, Core) - Integrations via HACS, editing YAML configuration files - Automation structure: Triggers, Conditions, Actions
W12	**Smart Lighting & Curtain**	2 Hrs	1.0	Relay modules, dimmers, motor limit settings, wireless control.	- Dimming types: Leading edge, Trailing edge, 0-10V, DALI - Curtain/Blinds: 4-wire dry contact controls vs RF vs Smart motors - Setting limit positions on curtain tracks (calibration)
W12/13	**Smart Irrigation & Energy Monitoring**	2 Hrs	1.0	Solenoid valves, current clamps, energy analytics dashboards.	- Solenoid valves: 24VAC vs 12VDC pulse controls, relay control - Non-invasive CT (Current Transformer) clamps installation & math - Configuring energy cost parameters, phase load balancing
W13	**Solar CCTV & IoT Fundamentals**	2 Hrs	1.0	Solar panel wattage/battery capacity calculations, 4G industrial routers.	- Solar panel types (Mono vs Poly), MPPT vs PWM charge controllers - Battery chemistries: LiFePO4 vs Lead Acid (depth of discharge) - 4G router configuration: SIM APN settings, watchdog timers

Wk	Topic	Hrs	Sess	Focus & Practical Scope	Theory PPT Points & Core Explanations
W13	**Site Survey & BOQ/Quotation**	3 Hrs	1.5	Site assessment checklists, creating a detailed Bill of Quantities (BOQ).	- Site Survey checklist: power source locations, wall types, line-of-sight - Structuring a Bill of Quantities (BOQ) with margins - Drafting project quotation templates & terms of payment
W13	**Project Planning & Installation Standards**	3 Hrs	1.5	Conduit paths, mounting heights, labeling and documentation.	- Conduit routing standards: separation of high/low voltage lines - Standard mounting heights (switches at 1.2m, DBs at 1.5m) - Labeling schemas for cables (ferrule tags, network rack indexing)
W13/14	**Testing, Commissioning & AMC**	2 Hrs	1.0	Handover checklists, Annual Maintenance Contract (AMC) structures.	- Loop testing, resistance logs, final load tests - Handover checklist elements: customer training, login transfers - Structuring AMCs: Comprehensive (parts included) vs Non-Comprehensive
W14	**Customer Support & Sales/Negotiation**	3 Hrs	1.5	Objections handling, customer support SLA design, pricing strategy.	- Service Level Agreements (SLA): Response time vs Resolution time - Handling sales objections (price objection, brand comparisons) - Value-based pricing vs cost-plus pricing strategies
W14	**Digital Marketing & GBP**	2 Hrs	1.0	Google Business Profile optimization, local SEO, online reviews.	- GBP: Setup, verify location, service list, geo-tagged photos - Local SEO: local keyword placement, citation management - Review generation strategies: QR codes on invoices, automated SMS
W14	**WhatsApp Business & Automation**	2 Hrs	1.0	Catalog setups, automated quick replies, broadcasts.	- Catalog setup: product images, descriptions, direct pay links - Automations: greeting messages, away messages, quick reply shortcuts - WhatsApp Business API vs standard App: limitations & rules
W14	**Business Setup & Entrepreneurship**	2 Hrs	1.0	Licensing, distributor network setup, credit flow, legal steps.	- Company registration types (Proprietorship, LLC/Partnership) - Vendor/Distributor registrations (distributor pricing tiers) - Working capital management, inventory turnover, tax filings

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Professional Practical Labs (20 Hours)

Wk	Lab Name	Hrs	Sess	Target Lab Activities	Theory PPT Points & Core Explanations
W15	**Network Rack Setup & Fiber Splicing**	2 Hrs	1.0	Cable dressing, patch panel punch-down, fusion splicing fiber.	- Rack units (U) standards, patch cords layout, heat dissipation - Fiber splicing safety: glass shards hazard, alcohol cleaning - Splicing: fiber stripping, cleaving (flat face), fusion alignment

Wk	Lab Name	Hrs	Sess	Target Lab Activities	Theory PPT Points & Core Explanations
W15	**Complete Office CCTV Installation**	4 Hrs	2.0	Mount cameras, route CAT6/Coaxial, configure NVR and remote P2P app.	- Reviewing site blueprints for camera placement validation - IP addressing scheme layout for surveillance VLAN - Configuring alerts, continuous/smart recording schedules
W15	**Villa Smart Home Installation**	4 Hrs	2.0	Wire smart switches, integrate curtain motor, set up Home Assistant hub.	- Wiring switch boxes with and without neutral wires safely - Calibrating curtain motor endpoints via RF/control bus - Dashboard layout best-practices for residential tablets
W16	**Access Control & Video Door Phone Setup**	3 Hrs	1.5	Wire reader to lock controller, configure EM lock power, mount VDP outdoor unit.	- Relay topologies: fail-safe locks wired to NC contacts - Connecting emergency push buttons (break glass) to cut power directly - Card scanning logic, authorization tables upload
W16	**Gate Automation & Boom Barrier Setup**	3 Hrs	1.5	Align barrier arm, connect induction loop, configure motor control board.	- Force limit adjustments on sliding gate motors for collision safety - Calibrating loop sensitivity parameters to detect cars not carts - Troubleshooting gate obstacle-detection photocells
W16	**Fire & Intruder Alarm Installation**	2 Hrs	1.0	Wire conventional fire panel, configure wireless intruder zones, pair keyfobs.	- End of line (EOL) resistor values placement in zones (Class B) - Programming delay zones (entry/exit paths) vs instant alarm zones - Sirens sounder limits, silent test procedures
W16	**Troubleshooting Lab**	2 Hrs	1.0	Find and resolve 3 student-hidden hardware/network/power faults on live setups.	- Diagnostic flowcharts: Power checks first, then continuity, then logic - Troubleshooting checklist templates usage - Reporting diagnostics and corrective action logs

3. Recommended Weekly Schedule Template

To maximize hands-on capability, each weekly topic is split across a 5-day progression. The daily 2-hour session is structured to minimize passive listening and maximize physical interaction with hardware.

Daily Session Time-Block Allocation

Within each 2-hour (120 minutes) daily session, manage time using the following split:

Phase	Duration	Focus Area	Key Activities
1. Concept & Demo	30 Minutes	Instructor-led briefing	Explaining core concepts, safety standards, and reviewing real-world wiring diagrams.

Phase	Duration	Focus Area	Key Activities
2. Active Lab Time	75 Minutes	Hands-on student execution	Wiring physical devices, configuring software settings, and running integration tests.
3. Review & Reset	15 Minutes	Verification & lab hygiene	Checking connections, backing up configurations, cleaning the workbench, and Q&A.

5-Day Weekly Rotation

Each weekly topic should follow this progression:

Day	Focus	Session breakdown	Expected Student Output
Day 1	**Concept & Wiring Topology**	- **30 mins:** Diagram explanation, spec sheets. - **75 mins:** Unboxing, identification, physical mounting on workbench. - **15 mins:** Component safety check.	Accurate block diagram drawn in lab book; components physically set up.
Day 2	**Configuration & Setup**	- **30 mins:** Software parameters & programming theory. - **75 mins:** Setting IPs, configuring software, testing initial controls. - **15 mins:** Configurations backed up.	Configured individual device accessible via control terminal.
Day 3	**Hardware Deployment & Cabling**	- **30 mins:** Termination standards (RJ45, RS485, Coaxial). - **75 mins:** Crimping, routing, powering through POE/SMPS. - **15 mins:** Multimeter voltage testing.	Fully wired and powered subsystem.
Day 4	**System Integration**	- **30 mins:** Relays, dry/wet contacts, trigger logic schemas. - **75 mins:** Interfacing subsystems (e.g. Video Door Phone triggering EM Lock). - **15 mins:** End-to-end flow validation.	Multi-device functional system executing logical commands.
Day 5	**Troubleshooting Lab & Assessment**	- **15 mins:** Fault-finding checklist methodology. - **90 mins:** Instructor-injected hardware/software fault resolution. - **15 mins:** Clean workbench reset.	Student successfully finds and repairs 2-3 faults.

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4. Key Implementation Guidelines

[!IMPORTANT]
Theory to Practical Ratio (30:70)
Do not spend more than 35 minutes per day on slide presentations. Use physical devices on the workbench from Day 1.
[!TIP]
The Fault-Injection Method
The best way to build troubleshooting skills is for the instructor to secretly create problems (e.g., swapping Tx/Rx wires, setting static IP conflicts, cutting power supply channels) and having students resolve them using multimeters and software diagnostic tools.
[!NOTE]
BOQ & Survey Portfolio
By the end of Module 4, every student should have built at least 3 custom proposals (Villa Smart Home, Office CCTV, Warehouse Security) to present as their portfolio.